

Retrieval, document review, and grounding

Search finds candidate pages. Document review walks a queued set of those pages. Grounding decides whether a drafted date, identifier, or number is allowed to persist. Cost is dominated by resending this evidence on every subsequent model step.

Citations are filename + page. Chunk ids change on re-ingest. A hit that only names a file is a locator, not evidence. Cover sheets (`divider=true`) are locators too — Document chat keeps `search_documents` open so the following data page can be read.

1. Hybrid search (under the harness)

`searchReportDocuments` / `searchReportDocumentsDetailed` in `src/lib/attachments/retrieval.ts`. Document chat uses hybrid. Analytics chat is keyword-first at the tool layer. The report body is not chunk-indexed — use `read_section`.

`classifyRetrievalQuery(trimmed)` → identifiers vs semantic vs mixed

1. File / span route

filename + outline spans
before chunk search

2. Exact identifiers

`document_pages.identifiers` &&
legacy ILIKE on raw_text

3. English FTS

`to_tsvector(contextual_text)`
`websearch_to_tsquery`

If exact IDs already fill limit, or lexical fills limit with real matches
skip the query embedding. Latency is mostly tool rounds + embed, not HNSW.

4. Vector arm (768-d) fused with FTS, then collapse to best chunk per page

Quoted phrases stay intact. AND across concepts. OR only inside a phrase family (e.g. Media Fill).

Post-rank: match-centered excerpts · quote-over-visual · locator ranking. TOC / running-header ID lists are last.
Caps: 16 hits, 80 excludePages, 900-char excerpts. A TOC-only grep retries excluding those pages (`requirementIndex`).
File-set questions (how many files, which folder, PDF vs Word, ready vs ingesting) use `list_attachments` — not this search.

Tool result includes a ready citation: [filename, p. N] or [filename] if page missing

Saved fields convert those to numbered [n] plus a trailing Citations: list. Do not reject a draft that omitted a known page.

Figure 1. Hybrid retrieval already runs under chat. Do not replace agentic grep with Elasticsearch or a hosted vector DB.

2. Search loop (when the harness hides grep)

`searchLoopDirective` in `search-loop.ts`. Shared by Document and Analytics, with different “this was a real cited page” rules.

Event this turn	search_documents	ask_user (Document)
Cited hit, locate/read, or two empty greps	Hide (directive read)	Hide until a page is actually read
Every hit is a cover sheet (dividerHits)	Stay open	Hide until read_document_page
Tool returned keepSearchOpen or unsupported_facts	Stay open	—
Active document review	Stay open (review tools own the step)	—
Analytics: a real cited page	Hide for the rest of the turn	Never ask which page — search/scan and say found or not

Never quiz the page number. After any grep, Document chat hides ask_user until read_document_page. Outline locates; it does not unlock a quiz.

3. Document review state machine

Empty inventories and open-set drafts with distributed evidence use a page walk, not another grep. Tools: `start_document_review` → `continue_document_review` → `finish_document_review`. Phase lives on the in-turn `DocumentReviewSession`.



- start** `planReviewPages` queues objective-matching pages (floor 8). Nearby same-file pages only — no leftover pad to 2500. `REVIEW_PAGE_FETCH_CAP = 2500` is listing safety so one file cannot starve others. It is not a walk cap.
- continue** 8-wide extract pool in one call. Stop starting new batches after ~60s (`REVIEW_CONTINUE_BUDGET_MS`). Leave 20s of the 270s abort window for persist. Transcript-backed batches skip the extract LLM.
- finish** Returns a capped findings sample (60). Seeds the ledger with `reviewedEvidence` (every walked page). Rehydrate only `status=complete` and `truncated !== true`. `coverageKey = selected documents' page counts + |obj|`.
- already_complete** Same `|obj|`: just finished → do not reset into another walk. Fill with `edit_table`.

Figure 2. `prepareDocumentReviewStep` forces the tool for the current phase. A leftover qualification finish does not unlock Calibration.

Coverage identity

Piece	Why it exists
<code>coverageKey</code> includes <code> obj </code> :	A finished calibration walk must not satisfy monitoring / associated instruments.
<code>coverageKeySatisfiesObjective</code>	Rehydrate when attachments are unchanged <i>and</i> the objective still matches.
Floor-8 skip of the PRQR	Does not unlock <code>edit_table</code> . CSV-OQ headers while the evidence file was never queued is not a complete inventory.
Matching finish this turn	Hide review tools even if the table is still empty — that emptiness is why you reviewed. Next step is <code>edit_table</code> .
Explicit retrieval pushback	Skips rehydrate. Queued inventory keeps comprehensive (<code>planKeepsComprehensive</code>).

placeholder_fill ≠ page walk. A populated table that still has `<date>` / `<identifier>` is adaptive closed-set search (`placeholder-fill.ts`). Header-only seeded tables (blank data cells) are empty inventories and still need a matching finished review.

4. Citation ledger and fact gate

Hard facts (dates, identifiers, measured numbers) in `draft_field` / `propose_edit` / `edit_table` must match a quote on a page this turn retrieved. Empty ledger or pages with no served quotes **fail open**.

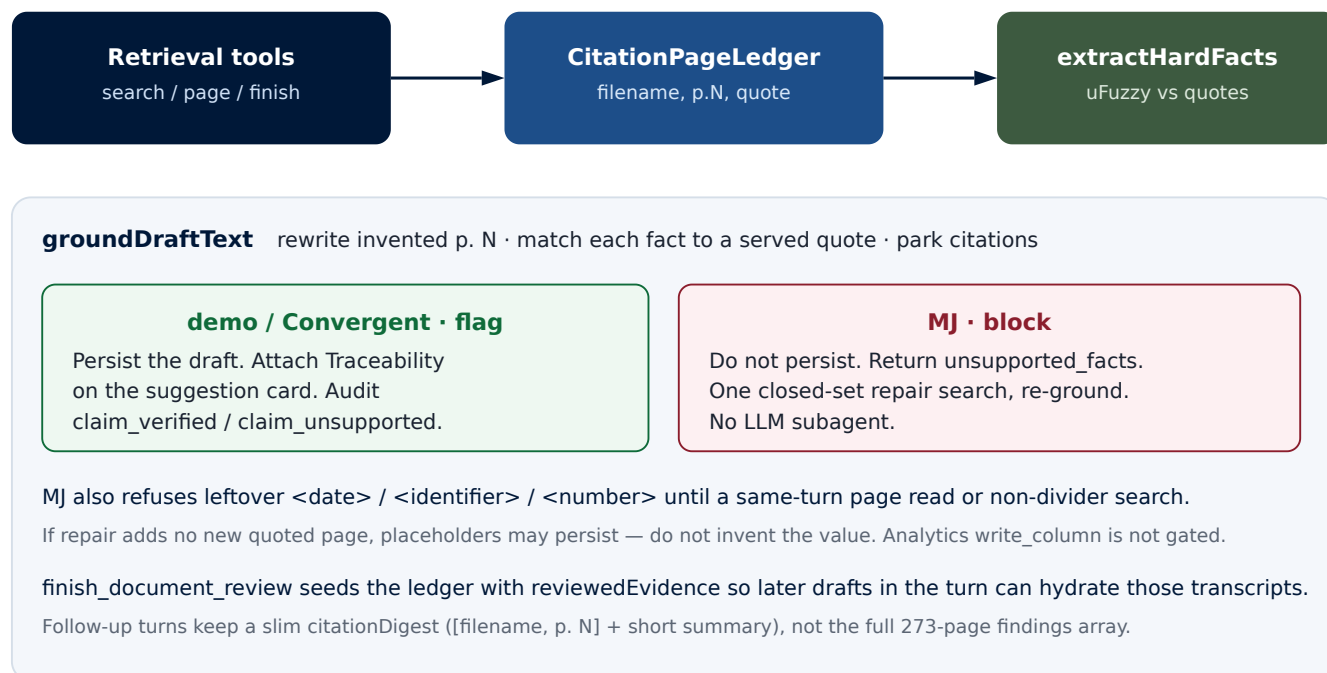


Figure 3. The model can still hallucinate a batch number. The harness is the last gate before an `ai_fix` comment is written.

What counts as a quote

- Search hit snippet (budget 900 chars)
- `read_document_page` transcript (12k chars)
- Every walked page on finish, not the 60-finding sample

What does not count

- The Documents index / `documentSummary`
- Filename-only `[file.pdf]` with no page
- Cover-sheet hits (`divider=true`)
- Prior-turn findings unless compacted into `citationDigest`

5. Compaction, budgets, abort

Every model step resends system prompt + tool JSON schemas + the growing tool-result transcript. A 273-page finish that keeps full findings will 500 the next message. That is why compaction exists.

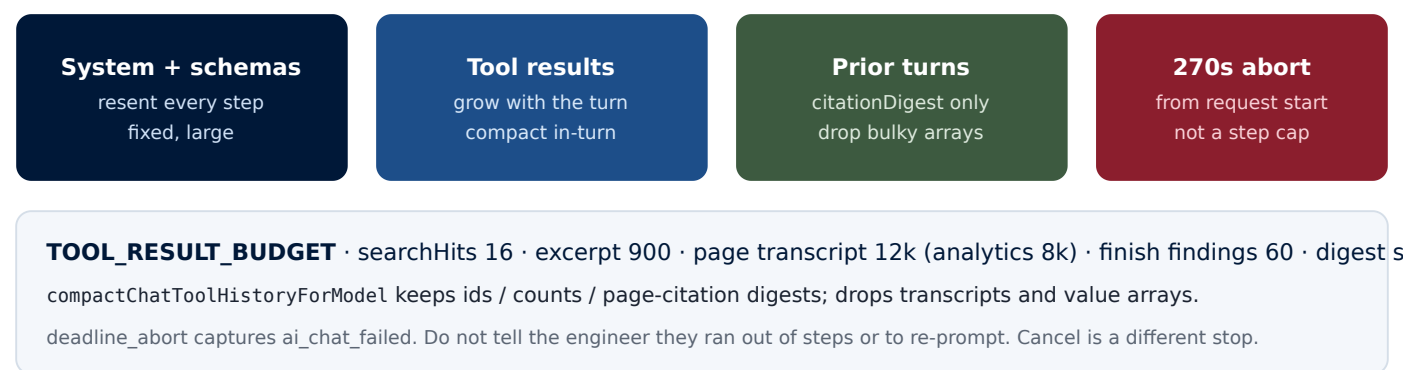


Figure 4. There is no per-turn tool-step cap. Loop guards live in prepareStep. Hobby function max duration must stay above 270s.

Constant	Value	File
CHAT_SERVER_ABORT_MS	270_000	assistant-turn.ts
REVIEW_CONTINUE_BUDGET_MS	60_000	document-review.ts
REVIEW_CONTINUE_DEADLINE_MARGIN_MS	20_000	leave room to persist after continue
REVIEW_EXTRACT_CONCURRENCY	8	in-flight extracts in one continue
REVIEW_OBJECTIVE_PAGE_FLOOR	8	review-page-plan.ts
REVIEW_PAGE_FETCH_CAP	2500	listing safety, not a walk cap

Debug a “it didn’t search” bug in this order. (1) Latest-turn intent — was it social? (2) ChatTurnPlan.retrievalPolicy — focused skips grep. (3) prepareReportChatStep activeTools on that step — search already hidden? (4) Langfuse generation — did the model see the tool? (5) Retrieval eval (pnpm retrieval-eval) if the page was wrong, not if the tool never ran.

6. Clicking a citation

In the document, chat, or suggestion Sources, [filename, p. N] (or numbered [n] mapped to a parked source) opens or reuses that attachment’s canvas tab and scrolls the PDF. Combined [file A, p. N, file B, p. M] splits into one link per file. Word preview cannot jump to a page. Missing or ambiguous filenames toast. Open canvases stay mounted while you switch to Report so the PDF does not re-parse.

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